

Public Economics

Lecture 5 · Budget Analysis and Deficit Financing

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Today's Questions

- Who actually decides what a government spends and taxes each year — and what stops that process from running an unlimited deficit?
- What does it even mean to “measure” a deficit correctly — nominal or real, cash or accrual, actual or cyclically adjusted?
- Do today's debts and deficits matter for the long run, and if so, why should we care?

Based on

Gruber, J. *Public Finance and Public Policy*, Ch. 4 (§4.1–4.5, Budget Analysis and Deficit Financing).

5.1 The Budget Process

5.2 Measuring the Budget's True Position

5.3 Do Debts and Deficits Matter? A Long-Run View

Deficits, Crowding Out, and Growth

Conclusion

5.1 The Budget Process

Budgets, Deficits, and Debt: The Basics

Three related concepts

Budget = planned spending and revenue for a fiscal year.

Deficit = spending $>$ revenue in a given year (a *flow*).

Debt = accumulated past deficits (a *stock*).

- The EU's **Maastricht rules** cap the deficit at 3% of GDP and gross debt at 60% of GDP for member states (Lecture 1).
- Every question in this lecture is really about *how* we measure these numbers, and *why* the numbers matter beyond the headline percentage.

Austria's Budget Cycle

1. The **Bundesministerium für Finanzen (BMF)** drafts the annual budget law (*Bundesfinanzgesetz*, BFG) and the medium-term expenditure framework (*Bundesfinanzrahmengesetz*, BFRG).
2. The Finance Minister opens the process with the **Budgetrede** — the budget speech — in the Nationalrat, followed by a first reading.
3. The **Budgetausschuss** (budget committee) reviews both drafts in detail, supported by the **Budgetdienst** — Parliament's own independent budget office.
4. A structured, multi-day plenary debate follows: general debate, then a chapter-by-chapter (*Untergliederung*) detail debate, then the final vote — by **simple majority**; the Bundesrat has no blocking role.
5. After the fiscal year ends, the **Rechnungshof** (Court of Audit) prepares the final accounts (*Bundesrechnungsabschluss*) for Nationalrat approval.

Source: Parlament Österreich, “Wie berät der Nationalrat das Budget?”; BMF.

Planning Four Years Ahead: The BFRG

- Since 2009, Austria's medium-term framework (**BFRG**) sets binding expenditure ceilings for a **rolling four-year horizon** — each year, a new fourth year is added.
- Since 2017, the BFRG is debated and decided *together* with the annual BFG each autumn.
- The 2009 reform also enabled genuine **two-year budgets** (*Doppelbudget*): used for 2013/14, 2018/19, and most recently **2025/26** (passed June 2025, alongside a BFRG running to 2029).

A rolling four-year window is Austria's answer to a problem every government budget faces:
today's decisions have consequences years beyond the current fiscal year.

Source: Parlament Österreich; BMF, Doppelbudget 2025/2026 press release (June 2025).

Der Fiskalrat: Austria's Fiscal Watchdog

- Austria's independent fiscal council traces back to the **Staatsschuldenausschuss** (1970); it was renamed and reconstituted as the **Fiskalrat** in **November 2013**.
- Institutionally housed at the **Oesterreichische Nationalbank (OeNB)** since 1997; a small dedicated office (around 6 staff) supports it.
- It is Austria's designated **Independent Fiscal Institution** under EU Regulation 473/2013: it monitors compliance with national and EU fiscal rules and publishes an annual macro-fiscal outlook.

Not quite the US Congressional Budget Office

The CBO (est. 1974, ~275 staff) formally *scores* pending legislation for Congress. The much smaller Fiskalrat does not score individual bills — it monitors and reports, closer to a fiscal watchdog than a legislative scorekeeper.

Source: Fiskalrat.at, “Organisation”; CBO, “10 Things to Know About CBO.”

Keeping Bund, Länder, and Gemeinden in Line

- Austria is a federal state: the Maastricht deficit and debt ceilings apply to *general* government — federal, state (*Länder*), and local (*Gemeinden*) combined.
- The **Österreichischer Stabilitätspakt (ÖStP 2012)**, in force since January 2012, is the domestic mechanism that allocates deficit and debt targets — and sanctions for missing them — across these three levels.

Does Austria have a constitutional “debt brake”?

No. Unlike Germany’s constitutionally entrenched *Schuldenbremse*, attempts to anchor one in Austria’s federal constitution failed to reach the required two-thirds majority — first in **December 2011**, then again in the Bundesrat in **October 2019**. Austria’s debt brake exists only as *ordinary law* (Art. 4 ÖStP 2012; §2(4) BHG 2013).

Source: Parlament Österreich, debate archive 7.12.2011; RIS, Österreichischer Stabilitätspakt 2012.

The EU Layer: Maastricht Rules, Reformed

- The original Maastricht reference values — **3% deficit-to-GDP, 60% debt-to-GDP** — are unchanged since the Treaty. What changed, in a reform that took effect **30 April 2024**, is how compliance is planned and enforced.
- The **net expenditure path** is now the single operational indicator: each member state agrees a multi-year fiscal-structural plan with the Council, tracked via a “control account.”
- The standard adjustment period is **4 years**, extendable to **7 years** if the country commits to specified reforms and investments.

Austria, right now

The EU Council opened an **excessive deficit procedure** against Austria on 8 July 2025, triggered by a 2024 deficit of 4.7% of GDP — with an approved path requiring the deficit below 3% by 2028.

Source: Consilium, “Council adopts reform of fiscal rules” (29 Apr 2024); European Parliament fact sheet.

Application: What Happens When a Fiscal Rule Is Relaxed?

Grembi, Nannicini & Troiano (2016) study a natural experiment in fiscal-rule design: Italy's 1999 national rule restricted municipal deficit growth, but was **relaxed in 2001** for municipalities under 5,000 residents.

- Comparing municipalities just above and below the 5,000-resident cutoff, the study finds that once freed from the restraint, municipalities let **taxes fall** and **deficits rise**.
- The effect was **larger** where the mayor could be re-elected, where more parties competed, and where voters were older — consistent with a political-economy story, not just a mechanical budget-constraint effect.

Fiscal rules like Austria's ÖStP or the EU's expenditure path are not just accounting devices — they bind real political incentives to spend and tax.

Source: Grembi, Nannicini & Troiano, "Do Fiscal Rules Matter?" *AEJ: Applied Economics* 8(3), 2016.

5.2 Measuring the Budget's True Position

Real vs. Nominal: The Inflation Tax

- Inflation erodes the *real* value of outstanding government debt — effectively an “inflation tax” on whoever holds that debt.
- Approximation: inflation tax (% of GDP) $\approx$ debt-to-GDP ratio $\times$ inflation rate.

Austria, end-2024 debt-to-GDP	79.9%
Austria, average 2024 inflation (HICP)	2.9%
Implied inflation tax	$\approx$ 2.3% of GDP

A deficit measured in purely nominal terms overstates how much the government's real debt burden actually grew.

Source: Statistik Austria, Öffentliche Finanzen 2025 (debt revision); Statistik Austria, VPI Jahr 2024.

Cyclical vs. Structural Deficits

- Deficits move **automatically** with the business cycle: recessions cut tax revenue and raise unemployment-related spending, even with no policy change (“automatic stabilizers”).
- To judge whether fiscal *policy* is loose or tight, economists strip out the cyclical component to get the **structural (cyclically adjusted) balance**.

Who computes this for Austria?

Both the **Fiskalrat** and the **European Commission** publish structural-balance estimates for Austria — the Commission’s version is the actual basis for EU fiscal-rule compliance, not the headline (nominal) deficit.

Source: Fiskalrat, Jahresbericht; European Commission, DG ECFIN.

Cash vs. Accrual Accounting: The International Experience

- **Cash accounting:** counts revenue/spending when money actually moves. Simple, but treats a euro spent on a durable investment (a bridge) the same as a euro spent on a one-off transfer.
- **Accrual (capital) accounting:** spreads the cost of long-lived investments over their useful life, like a firm's balance sheet — but is harder to implement and easier to manipulate through valuation assumptions.

International experience is mixed: **New Zealand** was the first country to fully implement accrual/capital budgeting (1989) and still uses it; **Denmark, Finland, and the Netherlands** adopted and later discontinued elements of it; **Britain** implemented it in 2001 and used it for its 2013 “Investing in Britain’s Future” plan. Source: Gruber, *Public Finance and Public Policy*, Ch. 4, §4.2.

Austria's Own Shift: The Haushaltsrechtsreform

Austria itself moved toward accrual-style budgeting at the federal level, in two stages:

- **Stage 1 (2009):** introduced the multi-year BFRG framework and enabled genuine two-year budgets.
- **Stage 2 (2013):** a wholly new Bundeshaushaltsgesetz (BHG 2013) — full *Wirkungsorientierte Haushaltsführung* (outcome-oriented budgeting, including mandatory gender budgeting), replacing pure cash accounting (*Kameralistik*) with **accrual double-entry bookkeeping** across three linked statements:

Statement	What it tracks
Ergebnishaushalt	Operating result (like a P&L)
Finanzierungshaushalt	Cash flow (the traditional cash position)
Vermögensrechnung	Balance sheet of assets and liabilities

Source: BMF, “Die Haushaltsrechtsreform des Bundes”; BHAG, Geschichte.

Static vs. Dynamic Scoring

- **Static scoring:** estimates a policy's budget effect holding the size of the economy fixed.
- **Dynamic scoring:** lets the estimate account for how the policy itself changes growth, employment, or behavior — and therefore future tax revenue.
- Dynamic estimates are usually *smaller* in absolute size than static ones for a tax cut, since some of the revenue loss is offset by higher growth — but they rest on more assumptions, and are correspondingly more contested.

In Austria, both Parliament's **Budgetdienst** and independent institutes (**WIFO**, **IHS**) routinely cost major policy proposals — the same static-vs-dynamic tension applies to their estimates.

Source: Gruber, *Public Finance and Public Policy*, Ch. 4, §4.2.

5.3 Do Debts and Deficits Matter? A Long-Run View

The Tool: Present Discounted Value

Present discounted value (PDV)

A euro received (or saved) t years from now is worth less than a euro today. To compare payments across time, discount each future flow by the interest rate r :

$$PDV = \frac{F_1}{(1+r)} + \frac{F_2}{(1+r)^2} + \cdots + \frac{F_T}{(1+r)^T}$$

Any government decision that trades an upfront cost for a stream of future costs or savings — building infrastructure, cutting taxes today and raising them later, borrowing now to invest — has to be evaluated this way, **not** by comparing raw undiscounted totals.

Worked Example: Should the Government Pave the Road?

A government can pave a highway with a new “wear-proof” material. Paving costs €4 billion *today*, but saves €400 million in maintenance costs at the end of *each* of the next 10 years.

$$PDV(\text{savings}) = \sum_{t=1}^{10} \frac{\text{€}0.4\text{bn}}{(1+r)^t}$$

Discount rate r	0%	4%	8%
PDV of savings	€4.00bn	€3.24bn	€2.68bn
Worth it (cost = €4bn)?	break-even	No	No

The politician's mistake

“It pays for itself in exactly 10 years, so it’s obviously worth it” ignores that the later savings are worth *less in today’s terms* than the euro spent now — at any positive interest rate, this project is a bad investment.

Why Current Debt Labels Can Mislead

- Governments make huge **implicit** promises — future pension and healthcare benefits — that never appear in the officially reported “debt” figure, which counts only explicit borrowing.
- Relabeling a payment (e.g. calling a benefit cut a “tax increase,” or vice versa) can leave the government’s *true* long-run fiscal position completely unchanged, even though the headline deficit number moves.

This is why economists supplement the official, cash-based debt figure with a much broader concept: the **present value of all future obligations**, explicit and implicit alike.

Source: Gruber, *Public Finance and Public Policy*, Ch. 4, §4.3.

Measuring Long-Run Sustainability: Austria's Pension System

The European Commission's **2024 Ageing Report** projects Austria's public pension expenditure from 2022 to 2070:

	2022	Peak (≈ 2032)	2070
Old-age & early pensions (% GDP)	11.4%	12.9%	—
Total public pension spending (% GDP)	13.7%	—	14.0%

- Spending rises as the baby-boomer generation retires (peaking around 2032), then eases back down — a net increase of only **~ 0.3 percentage points of GDP** over the full 48-year horizon.
- Domestically, the **Alterssicherungskommission** produces an annual 5-year assessment (due by 30 November) and a long-term report every three years, currently projecting to 2050.

Source: European Commission, 2024 Ageing Report — Austria Country Fiche; Sozialministerium, Alterssicherungskommission.

A Much Flatter Curve Than the US Story

Gruber's textbook centers this section on the US Social Security and Medicare trustees' long-run **fiscal imbalance** — a headline figure in the tens of trillions of dollars, driven by healthcare-cost growth as much as by aging itself.

Austria's trajectory looks very different

A ~ 0.3 -point rise in pension spending as a share of GDP over 48 years is a far flatter path than the US long-run imbalance. Two reasons: Austria's pension system was already reformed repeatedly (raising the effective retirement age, harmonizing civil-servant pensions); and — unlike the US case — this projection doesn't have to absorb open-ended healthcare-cost growth on the same scale.

The lesson generalizes: *how* a long-run projection is measured, and *which* obligations it includes, drives the headline number as much as demographics alone.

How Far Ahead Should We Look? Budget Windows

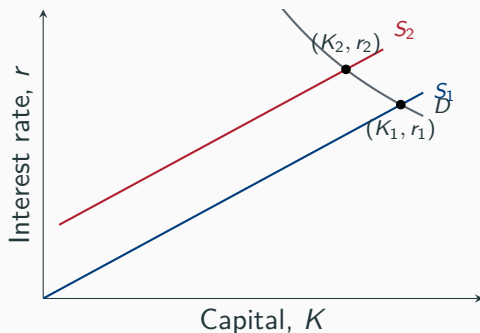
- A short budget window (one year) misses obligations that only bite later; a very long one (75 years, as US long-run scores sometimes use) is dominated by uncertain assumptions about growth, interest rates, and demographics.
- Austria's own answer sits in between on the *annual* planning side: the BFRG's rolling **4-year** window (§5.1) — extendable to **7 years** under the EU's reformed fiscal framework if reforms and investments are committed to.

There is no “correct” window — only a trade-off between capturing genuinely long-run obligations and trusting projections far enough out that the numbers stop being credible.

Source: Gruber, *Public Finance and Public Policy*, Ch. 4, §4.3; European Parliament, EU fiscal framework fact sheet.

Deficits, Crowding Out, and Growth

How a Deficit Can Crowd Out Private Investment



- Firms' demand for capital, D , meets the supply of private savings — a deficit means the state borrows from that same pool, which would otherwise fund private investment.
- This shifts the supply of loanable funds available to private borrowers left, from S_1 to S_2 .
- Equilibrium moves from (K_1, r_1) to (K_2, r_2) : interest rates **rise**, private capital investment **falls**.
- This is the standard **crowding-out** mechanism.

Source: Gruber, *Public Finance and Public Policy*, Ch. 4, §4.4 (Fig. 4-4).

Open Capital Markets Blunt Crowding Out

- The crowding-out story assumes savings are supplied *domestically* and inelastically. With fully open, integrated international capital markets, a country can borrow from the whole world's savings pool — so its own deficit barely moves the interest rate it faces.

This fits a small open economy like Austria especially well

As one country inside the euro area's single capital market, Austria effectively faces a supply of savings that is close to perfectly elastic — far more so than a large economy like the US, where domestic savings still matter enough that some crowding-out concern remains empirically credible.

The theoretical case for worrying about crowding-out is *weaker*, not stronger, for a small Eurozone member than for the textbook's own (US) example.

Source: Gruber, *Public Finance and Public Policy*, Ch. 4, §4.4.

The Secular Stagnation Puzzle

Real interest rates across the industrialized world have fallen sharply and persistently over the past four decades — and not just in one country:

Period	Real interest rate
Mid-to-late 1980s, average across industrialized nations	>5%
January 2000 (US)	4.3%
Pre-Covid, February 2020 (US)	−0.1%

Furman & Summers (2020) show this decline holds across Canada, France, Germany, Italy, Japan, the UK, and the US alike — a genuinely worldwide phenomenon, not a US-specific one.

Source: Furman & Summers (2020), cited in Gruber, *Public Finance and Public Policy*, Ch. 4, Fig. 4-5.

Austria and the Negative-Rate Era

- The ECB cut its deposit facility rate below zero in **June 2014** (to -0.10%), deepening it to -0.50% by **September 2019** — and rates stayed negative until **July 2022**.
- Through most of that period, Austrian (and German) long-term government bond yields traded at or below zero: investors were, in effect, *paying* the government to hold their savings.

This is the Eurozone's own lived version of the secular-stagnation puzzle — and the backdrop against which Austria's rising post-2020 debt ratio has to be judged.

Source: European Central Bank, key interest rate history.

The “New View” of Deficits

- If persistently low (or negative) real interest rates mean deficits no longer visibly crowd out private investment, some economists argue fiscal sustainability should be judged by the **ratio of interest payments to GDP**, not the debt-to-GDP ratio itself.
- This does **not** mean deficits are “free”: debts must still eventually be serviced, and the argument only strengthens the case for spending that pays for itself — productive investment with a return above the (low) cost of borrowing.

Why this matters for Austria’s debate

Austria’s Fiskalrat frames its warnings in the traditional debt-ratio terms (§5.1) — while the “new view” would ask instead how much of the deficit funds productive investment versus current consumption.

Source: Furman & Summers (2020), “Fiscal Sustainability of the United States”; Gruber, *Public Finance and Public Policy*, Ch. 4, §4.4.

Conclusion

Highlights

- A government budget is shaped as much by *process* — Austria's BFRG, Fiskalrat, ÖStP, and the EU's fiscal framework — as by economics alone.
- “The deficit” is not one number: nominal vs. real, cyclical vs. structural, and cash vs. accrual measures can all tell different stories about the same year.
- Present discounted value is the right tool for judging any policy that trades an upfront cost for a future stream — and it is exactly what long-run fiscal-sustainability measures (like pension projections) try to capture.
- The case for deficits crowding out investment is theoretically weaker for a small, financially open economy like Austria than for a large one — and persistently low real interest rates since the 1980s have made economists rethink how much to worry about debt levels at all.

Discussion Questions for Next Time

1. Austria's debt brake exists only as ordinary law, not constitutional law. Does that make it a meaningfully weaker commitment device — and if so, for whom?
2. The EU's reformed fiscal framework lets Austria take up to seven years to adjust, if it commits to reforms and investment. What kinds of “reforms and investment” should qualify, and who should judge that?
3. Grembi, Nannicini & Troiano found relaxed fiscal rules led Italian mayors to cut taxes and raise deficits, especially where re-election was possible. What does this suggest about designing fiscal rules for Austria's Länder and Gemeinden?
4. If persistently low real interest rates really do make the “new view” of deficits correct, should Austria be borrowing *more* to invest — and in what?
5. Austria's pension projections rise far less steeply than the US Social Security/Medicare story. Is that because Austria's demographic pressure is genuinely smaller, or because of how the obligation is defined and measured?

- Textbook: Gruber, J. *Public Finance and Public Policy*, Ch. 4 (§4.1–4.5, Budget Analysis and Deficit Financing), Worth Publishers.
- Grembi, V., Nannicini, T., & Troiano, U., “Do Fiscal Rules Matter?” *American Economic Journal: Applied Economics* 8(3), 2016.
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- Parlament Österreich: Häufige Fragen zum Budget; Wie berät der Nationalrat das Budget?; Doppelbudget 2025/2026.
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- European Commission, 2024 Ageing Report — Austria Country Fiche; Sozialministerium, Alterssicherungskommission.
- European Central Bank, key interest rate history.

Thank you!

Questions and discussion